Summary in Graph

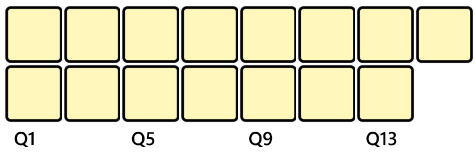
Exam Summary (GO Classes DA Test Series 2026 | Calculus and Optimization | Limit, Continuity | Topic Wise Test 1)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (10 + 5)
Total Marks:	**20** (10 + 10)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (76) out of 762 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the function $f(x) = \frac{4x^2 - kx}{x^2 + 12x + 32}$, where k is a constant.
Find the value of k for which $\lim_{x \to -4} f(x)$ exists.

A. $k = 16$
B. $k = -64$
C. $k = -16$

D. $k = 64$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 18sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If a function $f(x)$ is not defined at $x = a$, which of the following is a true statement?

A. $\lim_{x \to a} f(x)$ cannot exist.
B. $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$
C. $\lim_{x \to a} f(x)$ must approach infinity.
D. $\lim_{x \to a} f(x)$ might be equal to zero.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Determine the following limit:

$$\lim_{x \to \infty} \left(\sqrt{9x^2 + x} - 3x \right)$$

Which of the following is the correct evaluation of the limit?

A. $\frac{1}{3}$

B. 3

C. 0

D. $\frac{1}{6}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Determine the following limit:

$$\lim_{x \to 1^-} \left(\frac{2}{x} - \frac{1}{\ln(x)} \right)$$

Which of the following options correctly describes the limit?

A. The limit is $-\infty$.
B. The limit is finite and equal to 2 .
C. The limit is ∞.
D. The limit is 0 .

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

What values of m and b make the following function continuous:

$$f(x) = \begin{cases} x^2 - 7 & \text{if } x < -2 \\ mx + b & \text{if } -2 \leq x \leq 2 \\ 5 & \text{if } x > 2 \end{cases}$$

A. $m = 2, b = 1$
B. $m = 1, b = 2$
C. $m = 2, b = -1$
D. $m = 1, b = -2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

What is the value of $\lim_{x\to\infty}\left(\sqrt{x^2+x} - \sqrt{x^2-x}\right)$?

A. 0
B. 1
C. 2
D. ∞

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the function $h(x) = \frac{6x+5}{\sqrt{4x^2-9}}$. What is $\lim_{x\to-\infty} h(x)$?

A. $-\infty$

B. 0

C. 3

D. -3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

You are given the following information about the function $\mathrm{f(x)}$
(i) The domain of $f(x)$ is the interval $[-3, 1]$
(ii) The function $\mathrm{f(x)}$ is continuous in the intervals $[-3, -1)$ and $(-1, 1]$ and it is not continuous at $\mathrm{x} = -1$.
(iii) $f(-2) = -1.5$
using this information. Which, if any, of the following limits must exists?

A. $\lim_{x\to-2} f(x)$

B. $\lim_{x\to -2^+} f(x)$
C. $\lim_{x\to -1^+} f(x)$
D. $\lim_{x\to -1} f(x)$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

What is the value of $\lim_{x\to 6^-} (x-5)^{\frac{x-2}{x-6}}$?

A. 0

B. e^{-4}

C. e^{4}

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose $\lim_{x\to 6} |f(x)| = 2$. Which of the following statements must be true about $\lim_{x\to 6} f(x)$?

A. $\lim_{x\to 6} f(x)$ does not exist.
B. $\lim_{x\to 6} f(x) = 2$.
C. $\lim_{x\to 6} f(x)$ exists and is equal to either 2 or -2 .
D. There is not enough information about $f(x)$ to determine whether $\lim_{x\to 6} f(x)$ exists.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Suppose that a function f is defined by

$$f(x) = \begin{cases} 3x-2 & 0 < x < 4 \\ 18 & x = 4 \\ x^2 - 5x + 6 & x > 4 \end{cases}$$

Let $M = \lim_{x\to 4^-} f(x)$ and $N = \lim_{x\to 4^+} f(x)$. Find $3M + 5N$.

A. 12

B. 40

C. 56

D. 144

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the discontinuous function

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

and let $g(x) = x^2$. Compute $\lim_{x \to 0} f(g(x))$.
Which of the following is correct?

A. $\lim_{x \to 0} f(g(x)) = 0$
B. $\lim_{x \to 0} f(g(x)) = 1$
C. $\lim_{x \to 0} f(g(x))$ does not exist
D. $\lim_{x \to 0} f(g(x)) = -1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Evaluate the limit:

$$\lim_{x \to 0} \frac{x^2 - \cos(x) - \frac{1}{3}e^{3x} + \frac{4}{3}}{\ln(x+1)}$$

Select the correct answer:

A. -1
B. 0
C. 1
D. ∞

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $\lim_{x \to 1} \frac{a\sqrt{x+3} - b}{x-1} = \frac{1}{4}$, then what are the values of a and b ?

A. $a = 1; b = 2$
B. $a = 2; b = 1$
C. $a = 4; b = 2$
D. $a = 2; b = 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the following statements:
S1: If $f(x) \leq g(x) \leq h(x)$ for all x with $\lim_{x \to a} f(x) = -1$ and $\lim_{x \to a} h(x) = 1$, then $\lim_{x \to a} g(x)$ exists with $-1 \leq \lim_{x \to a} g(x) \leq 1$.
S2: Let f, g, h be functions with the same domain A and $f(x) \leq g(x) \leq h(x)$ for all $x \in A$. If $\lim_{x \to c} f(x) = L = \lim_{x \to c} h(x)$ for c some limit point of A, then $\lim_{x \to c} g(x) = L$ as well.
Which of the following options correctly interprets the provided statements?

A. Statement $\mathrm{S1}$ is true, and statement $\mathrm{S2}$ is false.
B. Statement $\mathrm{S1}$ is false, and statement $\mathrm{S2}$ is true.
C. Both statements $S1$ and $S2$ are true.
D. Both statements $S1$ and $S2$ are false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff